Multi-Armed Bandits

From planning to learning: regret, exploration vs. exploitation

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ESE 561 · Theory of Artificial Intelligence

From planning to learning

Remove one assumption, simplify everything else

MDP lecture: planning

- finite $\mathcal{S}$, $\mathcal{A}$; the state is observed
- **known model** $p(s'|s,a)$, $R(s,a,s')$
- value / policy iteration compute $\pi^\star(s)$

Today: learning

- **one state** s, M actions (arms) $a \in \{1,\dots,M\}$
- **unknown** reward distributions $p(r|a)$
- learn from the rewards collected *while playing*

At each step $t = 1,\dots,T$: pull an arm a_t, receive a reward $r_t \sim p(r \mid a_t)$.

In the MDP notation the reward after a_t was r_{t+1}; with a single state we index the pair (a_t, r_t) by the same t.

The bandit model

Assumptions

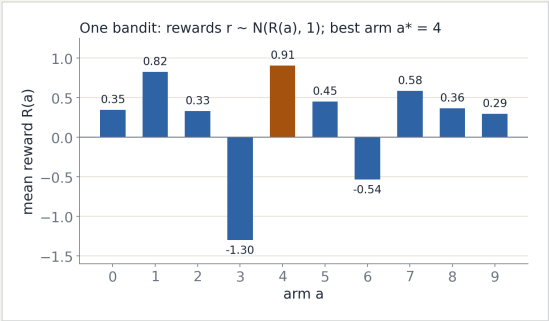
- $p(r|a) = p(r|s,a,s)$ unknown
- given the arm, pulls are i.i.d.
- $r \in [0,1]$ for the bounds
  (Gaussian, unit variance: the same)

Notation

$$R(a) = \mathbb{E}[r \mid a] \qquad \text{mean}$$

$$a^\star = \arg\max_a R(a) \qquad \text{best arm}$$

$$\Delta(a) = R(a^\star) - R(a) \geq 0 \qquad \text{gap}$$



10-armed testbed: $R(a) \sim \mathcal{N}(0,1)$, $r \sim \mathcal{N}(R(a),1)$

With R known the answer is trivial: $\pi(s) = a^\star$ forever. The difficulty is to learn $a^\star$ while playing.

An algorithm is a history-dependent policy

Before choosing a_t the agent has seen the history

$$h_{t-1} = (a_1, r_1, \ldots, a_{t-1}, r_{t-1}), \qquad h_0 = \varnothing.$$

An **algorithm** is a sequence of conditional distributions $\pi = (\pi_1, \pi_2, \ldots)$:

$$a_t \sim \pi_t(\cdot \mid h_{t-1}) \qquad \text{(deterministic: all mass on one arm)}.$$

Algorithm and environment define the distribution of the whole trajectory (chain rule):

$$P^\pi(a_1, r_1, \ldots, a_T, r_T) \;=\; \prod_{t=1}^{T} \underbrace{\pi_t(a_t \mid h_{t-1})}_{\text{agent}} \; \underbrace{p(r_t \mid a_t)}_{\text{environment}}$$

Compare $p(a, s', r|s) = \pi(a|s)\, p(s'|s, a)\, p(r|s, a, s')$ from the MDP lecture.
$\mathbb{E}^\pi$ = expectation under P^π: over the rewards *and* the agent's coin flips.

Maximizing reward = minimizing regret

Tower property, exactly as for $\mathbb{E}[r|s]$ in the MDP lecture:

$$\mathbb{E}^\pi[r_t] = \mathbb{E}^\pi\big[\,\mathbb{E}[r_t \mid h_{t-1}, a_t]\,\big] = \mathbb{E}^\pi[R(a_t)], \qquad \text{since } \mathbb{E}[r_t|h_{t-1}, a_t] = R(a_t).$$

Summing over t:

$$\mathbb{E}^\pi\Big[\sum_{t=1}^{T} r_t\Big] = \sum_{t=1}^{T} \mathbb{E}^\pi[R(a_t)] = T\,R(a^\star) - \mathbb{E}^\pi[R_T],$$

$$R_T \quad = \quad \sum_{t=1}^{T} \big(R(a^\star) \;-\; R(a_t)\big) \qquad \text{(regret)}$$

- what we lose, compared with an oracle who knows $a^\star$, because we have to learn
- random through the arms a_t; uses the means $R(a_t)$, not the noisy r_t
  (replacing $R(a_t)$ by r_t changes R_T but not its expectation)

What does "minimize over algorithms" mean?

For one *fixed, known* bandit the minimum is trivial: $\pi_t \equiv a^\star$ gives $R_T = 0$.

The algorithm does not know $p(r|a)$, so it must be good for **every** bandit ν in a class $\mathcal{E}$ (e.g. all reward distributions on $[0,1]$):

Worst case (minimax)

$$\inf_{\pi} \sup_{\nu \in \mathcal{E}} \mathbb{E}_\nu^\pi[R_T]$$

Bayesian, prior ρ over bandits

$$\inf_{\pi} \int \mathbb{E}_\nu^\pi[R_T]\, \rho(d\nu)$$

The bounds today ($T^{2/3}$, $\sqrt{T}$, $\log T/\Delta$) hold for every ν.
The notebook averages over bandits with $R(a) \sim \mathcal{N}(0,1)$: the Bayesian criterion.

Regret decomposition: only bad arms cost anything

Let $n_T(a)$ = number of pulls of arm a up to time T. Group the terms of R_T by arm:

$$R_T \;=\; \sum_{t=1}^{T} \Delta(a_t) \;=\; \sum_{a} \Delta(a)\, n_T(a), \qquad \mathbb{E}^{\pi}[R_T] \;=\; \sum_{a} \Delta(a)\, \mathbb{E}^{\pi}[n_T(a)]$$

Linear regret: $\mathbb{E}[R_T] \propto T$

some bad arm is pulled a fixed fraction of the time

Sublinear: $\mathbb{E}[R_T]/T \to 0$

eventually almost every pull is $a^\star$; goal $\sqrt{T}$ or $\log T$

Every regret bound today = a bound on $\mathbb{E}[n_T(a)]$ for each suboptimal arm.

Does using the history break the Markov property?

No. The Markov property is an assumption about the *environment*:

$$p(r_t \mid h_{t-1}, a_t) = p(r_t \mid a_t) \qquad \text{(used in } P^\pi \text{ and in } \mathbb{E}[r_t | h_{t-1}, a_t] = R(a_t)).$$

It says nothing about how the agent chooses its actions.

- MDP lecture: model known $\Rightarrow$ we *proved* that $\pi(s)$ is enough.
- Bandit: model unknown $\Rightarrow$ a stationary $\pi(s)$ pulls the same arm forever and never learns. A learning agent must use the past.

The environment is Markov in s; what changes over time is the agent's **knowledge**.

The knowledge state: just accumulate the rewards

Rewards of each arm are i.i.d., so their order does not matter. Before step t:

$$n_t(a) = \sum_{\tau<t} \mathbf{1}\{a_\tau = a\}, \qquad S_t(a) = \sum_{\tau<t} r_\tau \, \mathbf{1}\{a_\tau = a\}, \qquad \hat{R}_t(a) = \frac{S_t(a)}{n_t(a)}.$$

Bernoulli, or Gaussian with known variance: $(n_t(a), S_t(a))$ is a sufficient statistic.

Knowledge state $x_t = \big(t,\ n_t(\cdot),\ S_t(\cdot)\big)$, updated by

$$n_{t+1}(a_t) = n_t(a_t) + 1, \qquad S_{t+1}(a_t) = S_t(a_t) + r_t.$$

- every algorithm today chooses a_t from x_t (+ a coin flip for ε-greedy)
- given x_t, a_t, the next x_{t+1} depends only on r_t: (x_t) is Markov
- taken to the end: the posterior over $R(\cdot)$ is the state of a *belief MDP* (Gittins indices); beliefs return with POMDPs

Exploration vs. exploitation

Pull the arm that looks best, or learn more?

Exploit only

An unlucky first sample of $a^\star$ can lock the agent onto a worse arm forever.

Explore too much

Every pull of an arm already known to be bad costs its gap, again and again.

Your favorite restaurant, or a new one that might be better?

Five answers: **greedy** · **ε-greedy** · **Explore-First** · **UCB** · **Thompson sampling**

Greedy can get stuck forever

Pull every arm once, then

$$a_t = \arg\max_a \hat{R}_t(a) \qquad \text{(the MDP policy } \pi(s) = \arg\max_a R(a) \text{ with } R \text{ replaced by } \hat{R}\text{)}.$$

- $\hat{R}_t(a)$ is updated only for the arm that is pulled
- if the first reward of $a^\star$ is low, $a^\star$ is never pulled again, and $\hat{R}_t(a^\star)$ is never corrected
- if this happens with probability p:

$$\mathbb{E}[R_T] \quad \geq \quad p\,\Delta\,T \qquad \text{linear regret}$$

32%

of 200 random 10-armed bandits:
greedy ends on a wrong arm

ε-greedy deliberately randomizes the greedy policy

With probability $1-\varepsilon$ pull the greedy arm, with probability ε a uniformly random arm:

$$\pi_t(a \mid h_{t-1}) \quad = \quad \begin{cases} 1-\varepsilon+\dfrac{\varepsilon}{M}, & a = \arg\max_{a'} \hat{R}_t(a'), \\ \dfrac{\varepsilon}{M}, & \text{otherwise.} \end{cases}$$

Known model: randomizing loses reward

$$\mathbb{E}[r|s] \leq \max_a \sum_{s'} p(s'|s,a) R(s,a,s')$$

a statement about *exploitation*, when values are known

Unknown model: randomizing buys information

$P(a_t = a) \geq \varepsilon/M \Rightarrow n_t(a) \to \infty \Rightarrow$ $\hat{R}_t(a) \to R(a)$ (LLN) $\Rightarrow$ the greedy part finds $a^\star$. Stuck forever is impossible.

A constant ε never stops exploring

Even with perfect estimates, every step costs on average

$$\varepsilon \cdot \frac{1}{M} \sum_{a} \Delta(a) = \varepsilon \bar{\Delta} \qquad \Longrightarrow \qquad \mathbb{E}[R_T] \;\geq\; \varepsilon \bar{\Delta}\, T \quad \text{(linear)}.$$

0.153
predicted $\varepsilon \bar{\Delta}$
($\varepsilon = 0.1$, $\bar{\Delta} = 1.53$)

0.164
measured slope,
$t = 1000 \ldots 2000$

Remedies

- *decay* $\varepsilon_t \sim t^{-1/3}(M \log t)^{1/3}$: $\mathbb{E}[R_T] = O\big(T^{2/3}(M \log T)^{1/3}\big)$, no need to know T
  if the smallest gap is known, $\varepsilon_t = \min\{1, cM/(\Delta^2 t)\}$ gives $O(\log T)$
- *explore smarter*: ε-greedy tries a terrible arm as often as a nearly best one

Q-learning and DQN act ε-greedily; after learning the noise is switched off: deterministic again.

Explore-First: explore every arm n times, then commit

For $t \le nM$: round robin. Then

$$\hat{R}(a) = \frac{1}{n} \sum_{t \le nM:\ a_t = a} r_t, \qquad \hat{a} = \arg\max_a \hat{R}(a),$$

and pull $\hat{a}$ for the remaining $T - nM$ steps.

The estimate uses the observed rewards r_t; the means $R(a)$ are unknown.

The demo bandit, $M = 10$, $T = 1000$

n	committed to	regret
5	arm 0 (wrong)	566
10	arm 4 (best)	68
20	arm 4 (best)	136
50	arm 4 (best)	340

- too few pulls: commit to a wrong arm and pay its gap until T
  $n = 5$: gap 0.56 paid for 950 pulls
- too many: every exploration pull costs the average gap
  nM pulls at $\bar{\Delta} = 0.68$: 68, 136, 340

Hoeffding's inequality and the clean event

For n independent samples in $[0,1]$ with mean $R(a)$:

$$P\big(|\hat{R}(a) \;-\; R(a)| \;\;\geq\;\; \beta\big) \;\;\leq\;\; 2e^{-2n\beta^2}$$

Choose $\beta = \sqrt{2\log T/n}$: the right-hand side is $2e^{-4\log T} = 2/T^4$.

Clean event (union bound over M arms): with probability $\geq 1 - 2M/T^4$,

$$|\hat{R}(a) - R(a)| < \beta \quad \text{for all } a.$$

On the clean event the committed arm is almost the best:

$$R(\hat{a}) + \beta \;\geq\; \hat{R}(\hat{a}) \;\geq\; \hat{R}(a^\star) \;\geq\; R(a^\star) - \beta \quad\Longrightarrow\quad \Delta(\hat{a}) \leq 2\beta.$$

Explore-First: balancing the two costs gives $T^{2/3}$

With all gaps ≤ 1:

$$\mathbb{E}[R_T] \;\leq\; \underbrace{nM}_{\text{exploration}} \;+\; \underbrace{2T\sqrt{\tfrac{2\log T}{n}}}_{\text{commit to a slightly worse arm}} \;+\; \underbrace{T\cdot\tfrac{2M}{T^4}}_{\text{unclean event}}\;.$$

Minimize $f(n) = nM + 2T\sqrt{2\log T}\, n^{-1/2}$:

$$f'(n) = M - T\sqrt{2\log T}\, n^{-3/2} = 0 \quad\Longrightarrow\quad n^\star = \Big(\frac{T}{M}\Big)^{2/3}(2\log T)^{1/3}, \qquad f(n^\star) = 3Mn^\star.$$

$$\mathbb{E}[R_T] \quad = \quad O\big(T^{2/3}(M\log T)^{1/3}\big)$$

Weakness: n is fixed in advance and depends on T; exploration ignores what is observed.

Upper confidence bound

UCB: be optimistic about what you do not know

By Hoeffding, after $n_t(a)$ pulls, with probability $\geq 1 - 2/T^4$:

$$|\hat{R}_t(a) - R(a)| \leq \beta_t(a) = \sqrt{\frac{2 \log T}{n_t(a)}}.$$

Upper confidence bound and the rule (pull every arm once first):

$$\bar{R}_t(a) \;=\; \underbrace{\hat{R}_t(a)}_{\text{exploitation}} + \underbrace{\sqrt{\frac{2 \log T}{n_t(a)}}}_{\text{Hoeffding bonus}}, \qquad a_t \;=\; \arg\max_a \bar{R}_t(a)$$

- w.h.p. $\bar{R}_t(a) \geq R(a)$: an *optimistic* estimate
- rarely pulled arm $\Rightarrow$ large bonus $\Rightarrow$ try it: either it is good, or its bonus shrinks like $1/\sqrt{n_t(a)}$ and we learn it is bad
- deterministic given the history; explores only arms that might still be best

Why a bad arm is pulled only $O(\log T/\Delta^2)$ times

On the clean event, if UCB picks a suboptimal a:

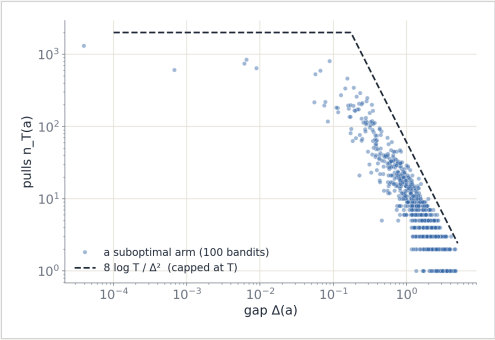
$$R(a) + 2\beta_t(a) \;\geq\; \bar{R}_t(a) \;\geq\; \bar{R}_t(a^\star) \;\geq\; R(a^\star)$$

(middle: the choice of a_t; outer: the confidence bounds)

Hence $\Delta(a) \leq 2\beta_t(a) = 2\sqrt{2\log T/n_t(a)}$:

$$n_t(a) \;\leq\; \frac{8\log T}{\Delta(a)^2}$$

After that many pulls the upper bound of a falls below $R(a^\star)$: UCB stops choosing it.



each dot: one suboptimal arm, 100 bandits, $T = 2000$

UCB regret: logarithmic in T, and that is optimal

Instance dependent. By the regret decomposition,

$$\mathbb{E}[R_T] \le \sum_{a:\,\Delta(a)>0} \Big(\frac{8 \log T}{\Delta(a)} + O(1) \Big) = O\Big(\sum_{a:\,\Delta(a)>0} \frac{\log T}{\Delta(a)} \Big).$$

Worst case. Tiny gaps make the bound blow up, but such arms are cheap. Split at δ:

$$\mathbb{E}[R_T] \le \underbrace{\delta T}_{\Delta(a)<\delta} + \underbrace{\frac{8M \log T}{\delta}}_{\Delta(a)\ge\delta} \xrightarrow{\delta=\sqrt{8M \log T / T}} \mathbb{E}[R_T] \le 2\sqrt{8\, MT \log T} = O\big(\sqrt{MT \log T}\big).$$

Lower bound (Lai and Robbins, 1985): for every reasonable algorithm

$$\mathbb{E}[n_T(a)] \gtrsim \frac{\log T}{\mathrm{KL}\big(p(\cdot|a),\, p(\cdot|a^\star)\big)}.$$

Anytime version: $\log t$ instead of $\log T$; $c\sqrt{\log t / n_t(a)}$ with $c = \sqrt{2}$ is Hoeffding, $c = 0$ is greedy.

Thompson sampling

Thompson sampling: sample a plausible world, act greedily in it

$$\tilde{R}(a) \;\sim\; p\big(R(a)\,\big|\,h_{t-1}\big) \;\text{ for every } a, \qquad a_t \;=\; \arg\max_a \tilde{R}(a)$$

Probability matching: $P(a_t = a \,|\, h_{t-1}) = P\big(a \text{ is the best arm} \,\big|\, h_{t-1}\big)$.

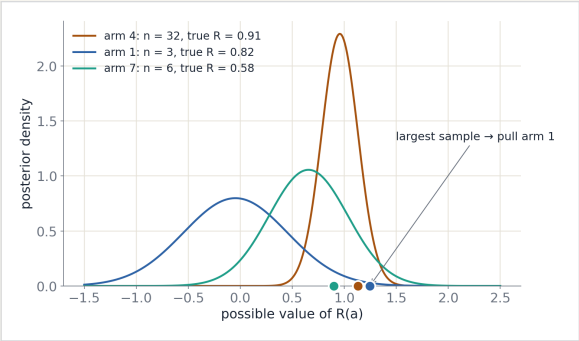
Gaussian testbed. Prior $R(a) \sim \mathcal{N}(0, \sigma_0^2)$, rewards $r \sim \mathcal{N}(R(a), 1)$. Bayes:

$$p\big(R \,|\, h_{t-1}\big) \;\propto\; e^{-R^2/2\sigma_0^2} \prod_{\tau<t:\, a_\tau = a} e^{-(r_\tau - R)^2/2} \;\propto\; \exp\Big(-\tfrac{1}{2}\big(\tfrac{1}{\sigma_0^2} + n_t(a)\big)R^2 + S_t(a)\,R\Big),$$

$$R(a)\,|\,h_{t-1} \;\sim\; \mathcal{N}\big(v_t(a)\,S_t(a),\; v_t(a)\big), \qquad v_t(a) = \frac{1}{1/\sigma_0^2 + n_t(a)}.$$

The posterior depends on the history only through $n_t(a)$, $S_t(a)$: the knowledge state again.

Thompson sampling explores arms that may still be best



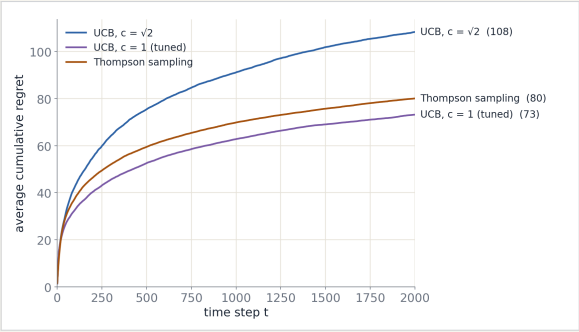
Demo bandit, after 60 steps:

- arm 4: pulled 32 times, narrow posterior
- arm 1 (true $R = 0.82$): 3 unlucky pulls, posterior mean ≈ -0.05 but wide
- its sample 1.25 beats arm 4's 1.13: try it again

Greedy would never touch arm 1 again.

ε-greedy randomizes blindly; UCB is deterministic and optimistic; Thompson randomizes in proportion to how plausible each arm is.

No knob to tune, close to a tuned UCB



200 bandits, $T = 2000$

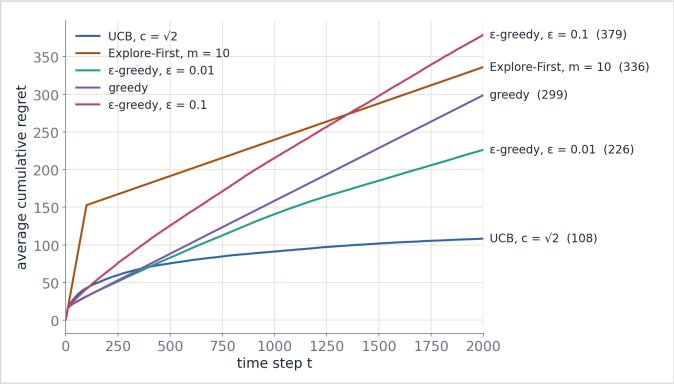
The prior width σ_0 plays the role of c. At $T = 1000$:

σ_0	regret
0.1	154 (= greedy)
0.3	127
1	68
3	75
10	74

Too narrow stops exploring; too wide only explores a bit longer. Theory: $O(\log T)$ for Bernoulli arms (Agrawal & Goyal 2012; Kaufmann et al. 2012).

Experiments

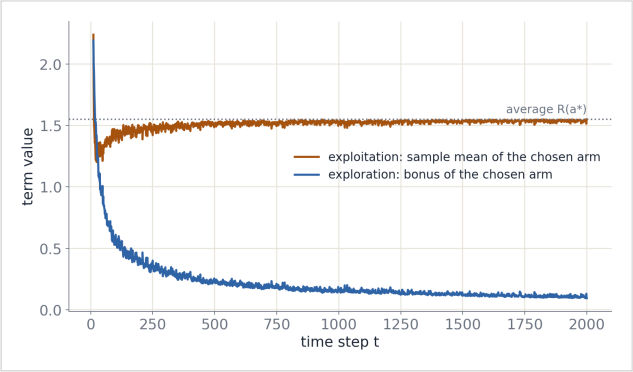
Straight lines are linear regret; UCB and Thompson bend



same 200 bandits for every algorithm

- **greedy**: a third of the runs stuck forever
- $\varepsilon = 0.1$: slope $\approx \varepsilon\bar{\Delta}$
- **Explore-First**: ramp of $nM = 100$ pulls, then wrong commits
- **UCB**: $\log t$
- Thompson (notebook): 80

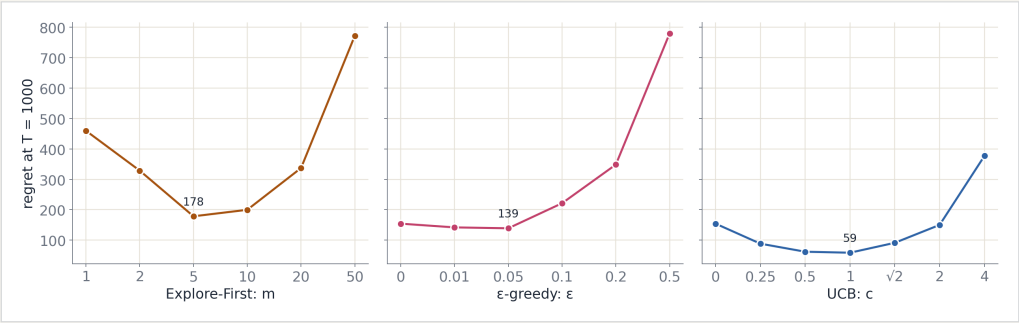
Inside UCB: nobody tells it to stop exploring



the two terms of the *chosen* arm, 200 bandits

- every pull raises $n_t(a)$ and shrinks its bonus; $\log t$ grows slowly
- the sample mean takes over and settles at $R(a^\star)$
- a bad arm returns only when $\log t$ has grown enough to lift its bonus

Every method has a knob, and every curve is U-shaped



- left end: too little exploration (stuck, wrong commits); right end: too much, paid at $\bar{\Delta}$
- UCB with $c = 0$ is greedy; the theoretical $c = \sqrt{2}$ is safe but conservative (best $c \approx 0.5$–1)
- the best n for Explore-First grows like $T^{2/3}$

Summary

Five ways to explore

algorithm	how it explores	$\mathbb{E}[R_T]$
greedy	never	linear, can get stuck
ε-greedy, fixed ε	uniformly at random, forever	linear, $\approx \varepsilon \bar{\Delta} T$
ε-greedy, $\varepsilon_t \sim t^{-1/3}$	uniformly, less and less	$O(T^{2/3})$
Explore-First	uniformly, first nM steps	$O\big(T^{2/3}(M \log T)^{1/3}\big)$
UCB	arms with large uncertainty	$O\big(\sum_a \frac{\log T}{\Delta(a)}\big)$, $O(\sqrt{MT \log T})$
Thompson sampling	as often as an arm may be best	$O(\log T)$ (Bernoulli)

Next: many states. ε-greedy becomes the behavior policy of Q-learning; UCB becomes the selection rule of Monte Carlo tree search (UCT).

In class: predict, then run the notebook

1. On what fraction of bandits does greedy get stuck? And with $\varepsilon = 0.01$?
2. Why does Explore-First with $n = 5$ lose 566 on the demo bandit, and $n = 50$ lose 340?
3. What plays the role of c in Thompson sampling? What does a too narrow prior do?
4. Which regret curves are straight lines, and which bend? Explain each from its algorithm.
5. Which arms does UCB pull a lot? Why are those pulls cheap?
6. Where is greedy on the UCB tuning curve? Is $c = \sqrt{2}$ the best value here?

Companion notebook: `Bandits_in_class.ipynb` (about 1–2 minutes to run end to end)